UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa
Department of Computing & Technology

AI vs the Brain: A Data Science Framework for Measuring AI Reliance and Human Engagement in Problem Solving

By

Richard Wambede
Bachelors of Science in Computer sciences and Information Technology

Reg No: *J25M19/ 005* , Student No: *B31328*
E-mail: *B31328@students.ucu.ac.ug*, Phone No: *0782386152*

A proposal submitted to the Faculty of Engineering Design & Technology for the study leading to the Degree in partial fulfillment of the requirements for the award of **Master of Science in Data Science and Analytics** of Uganda Christian University

June 2026

Declaration

I, *Richard Wambede*, declare to the best of my knowledge that the work in this proposal is original and has not been submitted before to any university/institution. It is the result of my effort.

Signature: ______________________________

Date: ______________________________

Ethical Generative AI Usage Declaration

Students may use generative artificial intelligence (AI) ethically as a "study buddy" or assistant rather than a replacement for critical thinking. This includes citing AI use and fact-checking all outputs to maintain academic integrity.

For purposes of this assignment, the following shall guidelines shall apply.

i. Generative AI may be used for brainstorming, and generating ideas to improve your work. You may also use AI for literature search, and summarizing journal papers for quick analysis and assessing relevance of publications. However, once the relevant publications have been identified, a student is expected to fully review the selected publications, identifying problems addressed, methods used, gaps identified and recommendations made.

ii. Where AI has been used as guided in (i) above, full disclosure of how it has been used should be made in the appendix B, including how AI was used, the AI tools (including mode versions), and a compilation on major AI prompts used.

iii. NO GENERATIVE AI CONTENT IS ALLOWED IN THE FINAL SUBMISSION. The final submission SHOULD be your original work.

A. **Declaring the non-use of AI for content generation**

I, *Richard Wambede*, declare that no content generated by AI technologies has been used in this proposal.

Signature: ______________________________ Date: ________________

B. **Acknowledging the use of AI as a study buddy**

Please refer to Appendix B for the complete declaration of AI usage for generating ideas, brainstorming, literature search, and summarizing only to improve your work.

Approval

This proposal has been submitted for examination with approval of the following advisors/supervisor.

Advisor: ***Dr. Francis Otto***
Department of Computing & Technology
Faculty of Engineering Design & Technology
Uganda Christian University

Signature: ______________________

Date: ______________________

Co-Advisor: ***Mr. John Bosco Wabwire***
Department of Computing & Technology
Faculty of Engineering Design & Technology
Uganda Christian University

Signature: ______________________

Date: ______________________

Abstract

Generative AI can improve speed and accuracy in problem solving, yet it can also shift cognition from human-generated intermediate reasoning to model-generated steps. The measurable risk is that higher AI reliance may reduce human engagement during problem solving and weaken future independent performance when AI is not available.

This proposal answers one central question: how does growing reliance on AI for cognitive tasks affect human problem solving, and does this reliance risk eroding essential cognitive skills. The core contribution is an end-to-end, reproducible measurement and modelling pipeline that:

1. Quantifies reliance behaviour as engineered features,
2. Constructs two auditable indices, Cognitive Offloading Index (COI) and Human Engagement Score (HES),
3. and lastly, estimates and predicts the impact of offloading on delayed, no-AI independent performance.

The study is a four-week quantitative research with a two-group, two-session design. In Session 1, a control group solves matched tasks without AI while an AI-assisted group solves the same tasks with ChatGPT under a strict protocol and behavioural logging. After 24 to 72 hours, all participants complete Session 2 with AI removed to measure delayed independent performance. The primary endpoint is Session 2 independent performance with AI removed, measured by task accuracy and rubric-scored independence and justification quality.

The analysis produces an evidence table with effect sizes and 95 percent confidence intervals, reliability statistics for scales and rubric scoring, and an exploratory cross-validated model that predicts Session 2 risk from Session 1 offloading features only. Deliverables include the cleaned dataset, a data dictionary, COI and HES scoring rules, code that reproduces all results, and a governance log that supports audit and defence.

> "The most dangerous mistake in the age of AI is believing that the system will think for us. Safeguarding our cognition is our shared responsibility; mine, yours, all of ours, beginning now."

Contents

Acknowledgments

Many thanks to my supervisors and Lecturers Dr. Terhemba M. Ahile, Dr. Kakuba Samuel, Dr. Bitalo, Dr. Innocent Ndibatya, among others for the guidance and useful feedbacks, as well as all my fellow students and participants at the Uganda Christian University. Special thanks to the Department of Computing & Technology for additional comments and guidance.

Chapter 1

Introduction

Executive Summary

Modern generative AI systems are not only reference tools. They can produce intermediate steps, draft reasoning and complete solutions, changing the human problem-solving pipeline itself. In data science terms, AI becomes a high-capacity external processor in the workflow, altering behaviour such as time allocation, verification frequency and the proportion of reasoning steps performed by the user.

This study positions AI reliance as a measurable behavioural construct rather than a vague attitude. Problem solving is treated as an observable sequence of actions, attempt, prompt, copy, verify, revise, that can be logged, transformed into features and modelled, then linked to downstream outcomes, especially independent performance when AI access is removed.

The central question is how growing reliance on AI for cognitive tasks affects human problem solving and whether it risks eroding essential cognitive skills. Brain here refers to cognitive performance and engagement measured through task outcomes, timing, verification behaviour, COI and HES.

The study will be grounded in the idea that human-AI collaboration will form hybrid cognitive systems, where the brain will offload some functions to AI, allowing focus on higher-order processes such as creativity and strategic thinking. However, this will create a paradox: increased reliance may reduce deliberate and effortful thinking.

To address this, the study applies data science methods to model the relationship between offloading and engagement through controlled experiments, using a mostly quantitative approach. The outcome is a practical framework that quantifies offloading and engagement, and estimates and predicts their impact on future independent performance under AI removal.

The resulting pipeline is designed for replication and audit, producing reusable scoring rules, a clean dataset and a decision-oriented evidence table. The study aims to show how AI can function not only as a performance tool but as a partner that strengthens human problem solving.

1.1 Background

Many recent AI discussions stop at one simple question, does AI help people finish tasks faster. That is not enough for a data science research degree. The real scientific question is how AI changes the way people think, and what that change does to later independent performance.

In this proposal, AI reliance is treated as measurable behaviour, not as an opinion. When a person uses AI to break down a problem, generate intermediate steps, or draft reasoning they would normally do themselves, that behaviour leaves traces. These traces include prompt counts, time spent in the tool, copying behaviour and verification actions. Those traces can be engineered into features. Features can be combined into indices like COI and HES. Indices and features can then be linked to later outcomes using effect estimation and prediction. That is the data science value of this study.

The motivation is also practical. In education and workplace tasks, AI may increase performance today while reducing independent capability tomorrow, if the user stops retrieving, elaborating and verifying on their own. This proposal therefore separates two outcomes. The first is immediate performance under AI access in Session 1. The second is delayed independent problem solving in Session 2 after AI is removed.

This study design provides a measurable test of whether reliance is acting as augmentation, where the user still reasons and verifies, or as substitution, where the model performs the reasoning and the user simply accepts it [1].

1.2 Problem Statement

Current generative AI use shifts problem solving from human-generated intermediate reasoning to AI-generated steps [1, 2, 3]. However, there is no lightweight, reproducible measurement and modelling pipeline in this context that can do three things. First, quantify offloading intensity as observable behaviour. Second, measure human engagement and verification behaviour during AI-assisted work [4, 5]. Third, estimate the effect of AI access on delayed independent problem solving when AI is removed [6, 7].

Technically, the problem is to build an experimental dataset and analytical framework that can estimate and predict how Session 1 reliance behaviour, COI, HES and verification, impacts Session 2 independent performance outcomes. The goal is to separate measurable augmentation, where AI support improves performance without reducing independent capability [1], from measurable substitution, where higher reliance leads to reduced independent performance after AI is removed [2, 8].

This distinction matters because it enables early detection of overreliance risk using data rather than assumptions [9, 10]. It is also the core technical need for responsible AI use in learning and knowledge work today.

1.3 Research Objectives

To understand how people depend on AI during problem solving, measure this behaviour using data science methods, and predict its impact on later independent performance [2, 8].

1.3.1 Specific Objective 1

To construct a labelled two-session dataset from a sample of university students and working adults in Kampala, engineer reliance and verification features from their behavioural logs, and compute COI and HES as auditable composite indices.

1.3.2 Specific Objective 2

To evaluate the effect of Session 1 AI reliance on delayed independent performance in Session 2 when AI is removed, using effect sizes, confidence intervals, and reliability checks [4, 5].

1.3.3 Specific Objective 3

To build and evaluate an exploratory prediction model that uses Session 1 features only to estimate Session 2 independence risk [8].

1.4 Research Questions

1.4.1 Research Question 1

How can we construct a good dataset, engineer useful features from raw behavioural logs, and compute COI and HES in a way that clearly shows AI reliance and verification behaviour [3]?

1.4.2 Research Question 2

How does Session 1 AI reliance affect delayed independent performance in Session 2 when AI is removed, and do COI and HES measure this relationship reliably [6, 7]?

1.4.3 Research Question 3

Can Session 1 features alone be used to predict Session 2 independent performance, and how well does such a model work as an early warning tool for overreliance risk [8]?

1.5 Significance of the Study

This study will turn a common worry into measurable data. Many people say AI can make people think less, but there is little proof from controlled studies with delayed testing. This study will measure AI use through behaviour logs and the Cognitive Offloading Index (COI), and measure engagement through the Human Engagement Score (HES). Both scores are scaled between 0 and 1, so behaviour can be compared fairly across different users [2].

This study will also test what happens when AI is removed. Most existing studies only check speed and accuracy while AI is present [11]. In this study, Session 2 removes AI for everyone and measures independent performance directly. For example, a drop of 10 to 30 percent in accuracy after AI removal may show a possible skill loss [6].

The study will also produce a reusable data science pipeline. It will include a clean dataset, a data dictionary, and clear formulas for COI and HES. This makes it easy for other researchers to repeat the study and check the results [9].

The study can help detect early signs of overreliance. If high COI scores, for example above 0.7, are linked to lower performance later, then COI can work as a warning signal. This idea is supported by research on AI trust and overreliance [4].

The study will also support better AI use rules. For example, simple actions like checking answers or explaining results before accepting them can be measured by counting how many times a user verifies a solution. This supports safer AI use based on real behaviour, not opinions [10].

Finally, this study will support future research. It will include simple checks like confidence intervals and basic tests to show if results are reliable. This will help other researchers build on this work and test it on larger groups [8].

1.6 Limitations

This study will be time bounded. The sample size, task types, and short delay period will limit how far the results can be generalised. The study will measure engagement and offloading using combined scores, COI and HES, so more testing on different tasks and groups will still be needed.

Future work should run a larger study with more task types and include one or two control conditions. This will help check if COI can be reduced without reducing performance, and if independent performance improves after AI is removed. This also connects to research on appropriate trust, where the goal is balanced use of AI, not overuse or rejection [12, 5].

The study suggests one simple rule for safe AI use. AI should not be judged only by speed and accuracy in the moment. It is also important to check what happens when AI is removed. For example, if COI increases above 0.7 and independent performance drops by 10 percent or more, this may show a risk to skills. This supports the need for designs that make users think first, since such methods have been shown to reduce overreliance in AI-assisted decisions [4].

For real-world use, risk control should follow existing AI guidelines. These include frameworks for safe and responsible AI use in education and decision making [9, 10].

In summary, this study will provide measurable evidence. AI can improve performance in the short term, but it may also increase offloading and reduce independent ability when removed. Using COI and HES together, and testing delayed performance, gives a simple way to separate useful support from risky reliance.

- The pilot sample size will limit generalisation. The study will focus on estimating effect sizes, not full population results.
- The short delay period will test near-term independence, not long-term skill change.
- Results may depend on task type and this will be reported clearly.
- Behaviour logs may be incomplete if participants do not follow instructions, but simple logging rules will reduce this risk.

1.7 Scope

This study will focus on how people use AI during problem solving and what happens when AI is later removed. The study will use a two-session design. In Session 1, some participants will use AI while others will not. In Session 2, all participants will solve tasks without AI. The goal is to compare behaviour and performance across the two sessions.

The study will focus on short-term effects, with a delay of about 24 to 72 hours between sessions. It will not measure long-term learning over weeks or months. The tasks will be limited to structured problem-solving activities, so results may not fully apply to all real-world situations.

The study will measure behaviour using logs such as prompts, time spent, copying and checking actions. These will be converted into features and combined into two main scores, the Cognitive Offloading Index (COI) and the Human Engagement Score (HES). The study will also build simple prediction models to link Session 1 behaviour to Session 2 outcomes.

The study will not focus on emotional, social or clinical aspects of cognition. It will only measure observable behaviour and task performance. The aim is to provide a clear and reproducible data science framework for analysing AI reliance [1, 2].

1.8 Limitations

This study will be limited by time, sample size, and task design. The number of participants will be small, so the results may not represent a larger population. The study will mainly estimate patterns and effect sizes, not final conclusions.

The delay between Session 1 and Session 2 will be short. This means the study will only measure short-term independence, not long-term skill development. Longer studies would be needed to understand lasting effects.

The study will use specific task types, so results may change if different tasks are used. Some tasks may require more reasoning, while others may rely more on memory or speed. This may affect COI and HES values.

Behaviour logs may not always be complete. Some participants may not follow instructions fully. This can affect the accuracy of the features. Simple logging rules and reminders will be used to reduce this problem.

The study will use combined scores (COI and HES), which may not capture all aspects of thinking. These scores will need further testing and validation in other studies [3, 8].

Finally, the study will not control all external factors such as prior knowledge, fatigue, or motivation. These may also affect performance and behaviour.

1.9 Key Definitions

- **Artificial Intelligence (AI):** AI refers to computer systems that can produce text, reasoning steps and solutions to tasks. In this study, AI acts as an external support tool during problem solving [11].
- **Cognitive Offloading:** Cognitive offloading is when a person uses an external tool to reduce their mental effort. In this study, it means shifting reasoning or memory tasks from the person to the AI system [6, 7].
- **Human Engagement:** Human engagement is the level of active thinking, checking and reasoning a person does during a task. It includes behaviours like verifying answers, explaining steps and revising solutions.

- **Cognitive Offloading Index (COI):** COI is a score that measures how much a person relies on AI during problem solving. It is built from features like AI use ratio, time in the tool, copy rate and low verification. Higher values, for example above 0.7, indicate stronger reliance on AI-generated steps.
- **Human Engagement Score (HES):** HES is a score that measures how actively a person participates in problem solving. It is built from engagement self-report, justification quality and low delegation to AI. Higher values indicate more independent thinking and verification.
- **Augmentation:** Augmentation means AI supports the person without reducing their ability to think on their own. The person still checks and understands the solution [1].
- **Substitution:** Substitution means AI replaces the person's thinking process. The person depends on AI outputs without much verification or understanding of the steps involved.
- **Delayed Independent Performance:** This is how well a person performs tasks after AI is removed. It is measured in Session 2 using task accuracy and rubric scored independence and justification quality.
- **Overreliance:** Overreliance happens when a person depends too much on AI and reduces their own thinking effort. This can lead to weaker performance when AI is not available [4, 5].
- **Verification:** Verification is when a person actively checks an AI-supplied answer using their own reasoning, re-computation or an alternative method. It is one of the key behaviours measured in this study.
- **Behavioural Feature:** A behavioural feature is a measurable variable extracted from raw activity logs. Examples in this study include prompt count, time in tool, copy rate and verification actions. These are the building blocks of COI and HES.
- **Session 1:** Session 1 is the first problem solving session. The control group works without AI. The AI-assisted group works with AI under a controlled protocol. Data collected here is used to compute COI and HES.
- **Session 2:** Session 2 is the delayed independence test. AI is removed for everyone. Participants solve a new but similar set of tasks on their own. This session measures whether prior AI use affected independent problem solving.
- **Feature Engineering:** Feature engineering is the process of turning raw log data into meaningful measurable variables. In this study, raw logs like prompt counts and time stamps are transformed into features that feed into COI and HES [1].
- **Prediction Model:** A prediction model is a statistical or machine learning tool that uses known inputs to estimate an unknown output. In this study, Session 1 features are used to predict Session 2 independence risk [8].
- **Effect Size:** Effect size is a number that shows how big a difference is between two groups. In this study, Cohen's d is used to report the size of differences between the AI group and the control group [13].
- **Confidence Interval:** A confidence interval is a range of values that is likely to contain the true result. This study uses 95 percent confidence intervals to show the uncertainty around each estimate.

- **Cross-validation:** Cross-validation is a method for testing how well a prediction model works on new data. It splits the dataset into parts, trains the model on some parts and tests it on the rest. This study uses grouped cross-validation by participant ID to avoid data leakage [8].
- **Data Leakage:** Data leakage happens when information from the test set accidentally enters the training set, making a model look better than it really is. This study prevents leakage by using only Session 1 features to predict Session 2 outcomes.

Chapter 2

Literature Review

2.1 Introduction

This chapter looks at existing research on generative AI, human thinking and problem solving. It focuses on how AI changes the way people engage with tasks, especially through cognitive offloading and verification behaviour. The goal is to understand what is already known, what is missing, and how this study will fill those gaps.

Recent studies show that generative AI is not just a tool for finding information. It actively changes how people think, solve problems and make decisions [1]. This makes it important to study both performance and behaviour, not just final task outcomes.

The chapter is organised into five areas. First, theoretical background. Second, cognitive offloading and human thinking. Third, generative AI and problem solving. Fourth, human-AI interaction and trust. Fifth, measurement of AI reliance and engagement. The chapter will end with a critical analysis and a summary of the three research gaps this study will address.

2.2 Theoretical Background

This study will be grounded in three overlapping theories. These are cognitive offloading, cognitive load, and human-AI collaboration.

Cognitive offloading is the use of external tools to reduce mental effort. In the past, this meant writing notes or using a calculator. Modern AI systems go further by performing reasoning steps, not just storing information [7, 3]. This changes what a person actually practises during problem solving.

Cognitive load theory explains how reducing unnecessary mental effort can improve performance. However, removing too much effort can also reduce learning, especially when key steps like retrieval and self-explanation are skipped [14]. The challenge will be finding the right balance between reducing unhelpful effort and keeping essential thinking steps active.

Human-AI collaboration is often described as hybrid intelligence, where humans and AI work together as a system. In this model, AI supports performance, but humans must still stay active in reasoning and verification [1]. When humans stop doing essential evaluation steps, the hybrid system becomes less safe, especially in high-stakes settings.

These three theories together explain why this study will measure both offloading and engagement. Offloading alone will not tell the full story. What will matter is whether the person stays actively involved while using AI, or whether they simply accept whatever the AI produces.

2.3 Cognitive Offloading and Human Thinking

Cognitive offloading has been studied widely in psychology. Research shows that offloading can improve immediate performance but may reduce memory and understanding over time [6, 7].

Modern AI increases the level of offloading by allowing users to delegate not only memory tasks but also reasoning steps. This changes what users practise during problem solving and may affect long-term independent capability [3].

Offloading is not always harmful. It depends on how it is used. If users return to the information, explain it in their own words, or verify it, learning can still happen. But if users simply accept AI outputs without checking, learning is likely to weaken [6].

However, the literature mostly discusses offloading at a general level. It does not clearly say which specific behaviours separate a person who is still thinking actively, augmentation, from a person who has handed all thinking to AI, substitution. This study will address that gap directly by measuring four specific behavioural features during problem solving.

The first feature is **AI use ratio**. This will be the proportion of tasks where the person requested AI assistance. A person who asks for help on every task will be showing stronger offloading than a person who only asks occasionally [5].

The second feature is **time in tool**. This will measure how much of the total task time was spent reading or interacting with AI output. A person spending most of their time in the tool will likely be delegating more of the thinking process [6].

The third feature is **copy rate**. This will be the proportion of the final answer that came directly from AI output. High copy rate will be a strong signal of substitution because the person will not be producing their own reasoning, they will be reproducing the AI's reasoning [4].

The fourth feature is **verification rate**. This will measure how often a person checked an AI-supplied answer using their own method, such as re-computing, estimating, or using an alternative approach. Low verification rate will mean the person accepted AI outputs without questioning them [4, 5].

Together these four features will form the building blocks of the Cognitive Offloading Index in this study. A person with high AI use ratio, high time in tool, high copy rate and low verification rate will be showing clear substitution behaviour. A person with low AI use ratio, low copy rate and high verification rate will be showing augmentation behaviour, even if they used AI at some point during the task [1].

This distinction is important because it will move the discussion from a vague question, did you use AI, to a specific measurable question, how did you use AI and did you still do your own thinking.

2.4 Generative AI and Problem Solving

Generative AI tools such as ChatGPT have been shown to improve task performance in many settings. Studies report higher accuracy and faster completion times when AI is used [11].

However, these benefits often come with reduced mental effort. Users may shift from active problem solving to passive acceptance of AI-generated answers. This creates a trade-off between performance now and understanding later.

Research also shows that doing tasks quickly with AI can give a false sense of learning. Completing a task fast does not always mean the person understands the process behind it. This is why delayed testing is important in this study. Session 2 removes AI and tests whether the person can still solve similar problems on their own [15].

Evidence from education shows a mixed picture. AI supported activities can improve performance while also reducing perceived effort. This may push people toward convenience-driven strategies if reflection is not built into the design [11]. Some students finish tasks faster with AI but perform worse when tested independently later, which is exactly the risk this study is designed to detect.

Research on creativity adds another concern. Generative AI can increase how much a person produces while reducing the variety of ideas across a group [16]. When AI becomes the default partner, solutions may start to look similar across users, which may signal reduced exploration and weaker independent reasoning over time.

2.5 Human-AI Interaction and Trust

Human-AI interaction research shows that trust is central but not enough on its own. Users must decide when to trust AI and when to check its output. However, research shows that trust attitudes alone do not reliably produce calibrated reliance, especially when users form simple habits like always following or always ignoring AI suggestions [5, 12].

Studies show that users often rely on simple rules such as always accepting or always ignoring AI outputs. This can lead to overreliance or underuse, neither of which is safe in settings where accuracy matters [5].

Controlled experiments show that standard explanations from AI systems do not reliably prevent overreliance when the AI is wrong, unless the interface forces deeper checking behaviour [4]. This reveals a design trade-off. Adding friction can reduce blind acceptance, but too much friction can reduce usability and push users to bypass safeguards altogether. The open challenge is to design friction that is measurable and calibrated, enough to increase verification without destroying efficiency gains.

Research on AI-generated content adds a further concern. Experiments show that people can struggle to tell AI-generated text from human text, and that synthetic content can be persuasive even when it is wrong [**?**]. This strengthens the case for verifiable checking routines rather than trust alone, which aligns with the emphasis on accountability and risk management in responsible AI guidance [9, 10].

Design solutions such as forcing users to explain answers or check results before submitting can reduce overreliance. However, too much control can reduce usability, so a balance is needed [4].

2.6 Measurement of AI Reliance and Engagement

A key challenge in the literature is how to measure AI reliance. Many studies rely on self-reported trust or attitudes, which do not always reflect actual behaviour [5]. A person may say they check AI outputs carefully, but their logs may show something different.

Recent research suggests that behavioural measures are more reliable than self-report alone. These include actions such as verification, copying, time spent in the tool and prompt usage [8, 2]. However, most studies measure these factors separately. There is no single combined index that captures both reliance and engagement together in one auditable framework.

This study will address that gap by combining four specific behavioural features into the Cognitive Offloading Index. These will be AI use ratio, time in tool, copy rate and verification rate. Each feature will capture a different aspect of how a person interacts with AI during problem solving [4, 5].

AI use ratio and time in tool will measure how much a person depended on AI. Copy rate will measure whether the person produced their own reasoning or simply reproduced the AI output. Verification rate will measure whether the person checked the AI output using their own thinking. Together these four features will separate a person who used AI as a support tool from a person who let AI do all the thinking [1].

Engagement will be measured separately through the Human Engagement Score. This will combine self-reported engagement items, justification quality scored by rubric, and low delegation to AI. The reason for keeping COI and HES separate is important. A person can have high engagement and still offload some steps to AI. These are not the same thing and should not be combined into one score [8].

Engagement will also be observed through task behaviour. A person who writes a detailed justification in their own words, checks their answer before submitting, and spends time on the task rather than in the AI tool will be showing high engagement regardless of whether they used AI or not [2].

Recent evidence links higher AI dependence to weaker critical thinking and fatigue-related mechanisms, while AI literacy can reduce some of these negative effects [8, 2]. However, a weakness in many of these findings is that they are correlational. They cannot show that AI use caused lower thinking. Weaker baseline skills could also drive heavier AI reliance. This is why this study will use a controlled design with two groups and a delayed no-AI session, where reliance will be manipulated and independent performance will be measured after AI is removed.

2.7 Comparative Analysis of Existing Studies

Study	Methodology	Dataset	Performance	Limitations
Grinschgl et al. (2021)	Controlled experiment	Memory tasks	Better task performance	Lower memory retention
Buçinca et al. (2021)	Experiment on AI decision tasks	Lab participants	Less overreliance with forcing functions	Limited real-world validity
Deng et al. (2024)	Meta-analysis of 50+ studies	Education datasets	Better performance, less effort	Mixed results across contexts
Gerlich (2025)	Survey and analysis	Adults using AI tools	Higher AI use linked to lower critical thinking	Correlational, no delayed test
Tian & Zhang (2025)	Survey study	University students	AI dependence linked to fatigue and weaker thinking	Self-reported data bias
Doshi & Hauser (2024)	Controlled experiment	Online workers	More individual output but less idea diversity	Lab context only
Spitale et al. (2023)	Experiment	Online participants	People struggle to detect AI-written content	Context specific results
Mehrotra et al. (2024)	Systematic review	Human-AI interaction studies	Trust attitudes do not reliably predict actual reliance behaviour	No behavioural measurement

Table 2.1: Comparison of selected studies on AI use, cognitive offloading and learning outcomes.

The table shows that most studies agree AI can improve speed and performance in the short term. However, there is much less agreement on its effect on learning and independent thinking. Many studies also lack behavioural measurement and delayed testing, which are both important for understanding what happens after AI is removed. This study will directly address both of these gaps.

2.8 Empirical Studies on AI and Learning Outcomes

Empirical studies show mixed results. AI often improves immediate performance, but its effect on learning is less clear [11].

Some studies show reduced mental effort and weaker critical thinking when AI is heavily used [2, 8]. Others show that AI can support learning when it is combined with reflection and active engagement. The difference seems to depend on whether the person re-processes AI output actively or simply accepts it passively.

Research also shows that AI can increase individual output but reduce diversity of ideas across a group. This suggests that users may rely on similar patterns generated by AI, which can signal reduced exploration and weaker independent reasoning over time [16].

One important finding from education research is that students who use AI heavily during practice tasks sometimes perform worse on independent tests later. This pattern supports the idea that AI use during learning can look like progress while actually reducing the depth of understanding [11, 15].

Another concern is that empirical findings in this area are mostly correlational. A student who uses AI more may already have weaker skills, which drives both the heavier AI use and the lower independent performance. This is why a controlled two-group design with a delayed no-AI session is needed to make stronger claims about the direction of the effect [6, 8].

2.9 Critical Analysis of Existing Studies

A key limitation in current research is the lack of behavioural measurement. Many studies focus on trust or perception rather than actual user actions during problem solving [5]. Asking someone whether they trust AI is very different from measuring what they actually do when AI gives them an answer.

Specifically, most studies do not measure the four behavioural features that this study will argue are central to detecting substitution versus augmentation. These are AI use ratio, time in tool, copy rate and verification rate. Without these features, it is impossible to say whether a person used AI as a support tool or let AI do all the thinking for them [4, 5].

For example, two people can both report using AI during a task. But one person may have asked one question, checked the answer independently and written their own justification. The other may have copied the AI output directly and submitted it without any checking. These two behaviours are very different, yet most studies would record both as simply AI use [1].

Another limitation is the focus on immediate performance. Few studies test what happens after AI is removed. This makes it difficult to measure true learning or independence [6, 7]. A person can perform well during a task with AI and then fail the same type of task one day later without AI. Without a delayed no-AI session, this risk will remain invisible.

There is also a lack of integrated measurement. Most studies measure offloading, engagement or performance separately rather than combining them into linked indices. This makes it hard to see the full picture of how AI changes the thinking process [8]. For instance, a person can be highly engaged and still offload heavily. These two things are not opposites, and measuring only one of them will give an incomplete picture.

Finally, many findings are based on correlation, which does not show causation. Controlled experiments with delayed testing and a no-AI condition will be needed to make stronger claims about the effects of AI use on independent problem solving [2, 11].

2.10 Research Gaps

Based on the reviewed literature, three main gaps are identified. Each gap directly motivates one part of this study.

Gap 1: Measurement Gap

Most existing studies do not provide an auditable, feature-based index that converts AI reliance into measurable variables. They either use self-report alone or measure only one behaviour at a time. Specifically, no existing study combines AI use ratio, time in tool, copy rate and verification rate into a single composite index that can separate substitution from augmentation [4, 5].

This study will address this gap by defining a fixed feature set for reliance and verification, then computing COI and HES using deterministic formulas backed by a data dictionary. This will mean another researcher can take the same raw logs and reproduce the exact same scores [1].

Gap 2: Delayed Endpoint Gap

Most studies only measure performance while AI is present. They do not test what happens after AI is removed. This means the risk of skill erosion is never directly observed, only assumed [6, 7].

This study will address this gap by using Session 2 with AI removed as the primary independence endpoint. If the AI-assisted group performs worse in Session 2 than the control group, that will be direct evidence of a short-term independence cost. If they perform similarly, that will also be useful evidence showing no measurable short-term cost for these tasks and contexts [11].

Gap 3: Validity Gap

Many studies report results without providing reliability evidence for their scoring tools. Rubric scores and composite indices can vary a lot depending on who scores them and how the formula is defined [8, 2].

This study will address this gap by using a written rubric with clear scoring dimensions, double scoring at least 20 percent of responses, and computing Cronbach's alpha for multi-item scales and ICC or kappa for rubric scoring. This will provide evidence that the measures used in this study are consistent and trustworthy [4].

2.11 Summary

This chapter has reviewed key studies on AI, cognitive offloading and human engagement. The findings show that AI can improve performance in the short term but may also change how people think and learn.

The review has shown that most existing studies discuss offloading and reliance at a general level. They do not identify the specific behavioural features that separate a person who is still thinking actively from a person who has handed all thinking to AI. This study will fill that gap by measuring four specific features during problem solving. These are AI use ratio, time in tool, copy rate and verification rate. Together these features will form the Cognitive Offloading Index and will make it possible to distinguish substitution from augmentation in a measurable and reproducible way [1, 4, 5].

The review has also shown that most studies do not test what happens after AI is removed. This study will address that by using Session 2 as a delayed independence endpoint where AI is removed for everyone [6, 7].

Finally, the review has shown that measurement tools in this area often lack reliability evidence. This study will address that by computing Cronbach's alpha for scales and ICC or kappa for rubric scoring on a double scored subset [2, 8].

Chapter 3

Methodology

3.1 Introduction

This chapter will explain how the study will be carried out from start to finish. It will describe the research design, participants, experimental procedure, variables, data processing and analysis methods. The aim is to provide a clear and practical plan that will produce measurable evidence on how AI use affects human thinking and independent problem solving.

The study will be designed to be simple but strong. It will focus on collecting clean data, building clear measures and producing results that can be tested and repeated by other researchers.

Each section in this chapter maps directly onto one of the three specific objectives of this study. The research design and participant selection will support Specific Objective 1. The variables, measurement and data processing sections will also support Specific Objective 1. The data analysis and statistical testing sections will support Specific Objective 2. The modelling and evaluation sections will support Specific Objective 3. This mapping will ensure that every methodological step has a clear purpose tied to the research questions.

3.2 Research Design

This study will use a quantitative, between-groups experimental design with a delayed follow-up session. This type of design is suitable for testing cause and effect relationships between variables [17, 18]. This section directly supports Specific Objective 1, which requires constructing a labelled two-session dataset.

Participants will be divided into two groups. Group A will be the control group, and participants will complete Session 1 without AI. Group B will be the AI-assisted group, and participants will complete Session 1 with ChatGPT under a fixed and controlled protocol.

The most important part of the design will be Session 2. After a delay of 24 to 72 hours, AI will be removed for all participants. They will then complete a new but comparable set of tasks. This session will act as the independence test. It will help measure whether participants can still solve problems on their own after using AI. This directly supports Specific Objective 2, which requires evaluating the effect of Session 1 AI reliance on delayed independent performance.

If the AI-assisted group performs worse in Session 2, this will suggest that heavy reliance on AI may reduce short-term independent thinking and problem solving [6, 7].

To reduce bias, both groups will complete two warm-up tasks without AI before Session 1. This will provide a baseline measure of ability. It will be used to check if the groups are balanced and to adjust the analysis if needed [19]. This baseline will also help account for differences among participants, since prior ability is one of the main sources of variation in this kind of study.

3.3 Conceptual Framework

This study will follow a simple but clear framework. AI use will be treated as the input. Behaviour during problem solving will be the process. Independent performance will be the outcome.

AI use in Session 1 will affect how participants behave. This behaviour will be measured using two indices, the Cognitive Offloading Index (COI) and the Human Engagement Score (HES). COI will capture how much thinking is shifted to AI. HES will capture how much the participant is actively thinking and verifying.

These behavioural measures will then influence the final outcome, which is performance in Session 2 when AI is removed. In simple terms, AI use will affect behaviour, and behaviour will affect performance.

This framework maps directly onto the three specific objectives of this study. Specific Objective 1 will cover the input and process parts of the framework, by constructing the dataset, engineering the behavioural features and computing COI and HES. Specific Objective 2 will cover the outcome part, by evaluating the effect of Session 1 reliance on Session 2 independent performance. Specific Objective 3 will use the full framework to build a prediction model that estimates Session 2 independence risk from Session 1 features alone [1, 2].

The framework also makes a clear distinction between augmentation and substitution. Augmentation will be observed when a participant uses AI but still reasons, verifies and produces their own justification. Substitution will be observed when a participant delegates most thinking to AI, copies outputs and does little or no verification. COI and HES together will make this distinction measurable and auditable [4, 5].

3.4 Study Setting and Participants

3.4.1 Target Population and Justification

This study will target university students and working adults aged 18 and above in Kampala, Uganda. More specifically, participants will be recruited from universities, vocational training institutes, workplace groups and ICT labs within Kampala.

This population was chosen for three reasons. First, university students and working adults regularly perform thinking tasks such as analysis, planning and decision making, which makes them a suitable group for studying AI reliance during problem solving. Second, this group is likely to have some familiarity with AI tools, which means their reliance behaviour will be more meaningful to measure than a group that has never used AI before. Third, recruiting from multiple sites within Kampala will increase variation in education level and AI familiarity, both of which will be recorded and used as covariates in the analysis [20].

This study will use convenience sampling because it is a short-term project. This approach is common in experimental research where quick data collection is needed [20]. The minimum target will be 30 to 40 completed cases for the pilot, with about 15 to 20 participants per group. This size is suitable for pilot studies that focus on feasibility and effect size estimation [21].

3.4.2 Inclusion Criteria

A participant must meet all of the following conditions.

- Be 18 years or older.
- Be able to read and write English well enough to complete the task prompts.
- Be available to complete both Session 1 and Session 2 within the study window.
- Provide informed consent before taking part.

3.4.3 Exclusion Criteria

A participant will be excluded if they meet any of the following conditions.

- Cannot commit to completing Session 2.
- Have already taken part in a previous round of this study.
- Cannot complete the tasks within the session time limit due to reading difficulty or non-compliance with instructions.

3.4.4 Accounting for Differences Among Participants

Prior knowledge, education level and AI familiarity are expected to vary across participants. These differences could affect both Session 1 performance and Session 2 independent performance. This study will account for these differences in three ways.

First, all participants will complete two warm-up tasks without AI before Session 1 begins. These warm-up tasks will provide a baseline ability score for each participant. This baseline will be used to check whether the two groups are balanced before the main analysis begins, and to adjust the regression model if the groups are not balanced [19].

Second, background variables will be recorded for every participant at the start of the study. These will include education level, self-rated AI familiarity on a scale of 1 to 5, and English comfort level. These variables will be used as covariates in the adjusted analysis to reduce the influence of pre-existing differences between participants.

Third, a task familiarity flag will be recorded for every task episode. This flag will ask whether the participant has seen a similar problem before. This will turn prior task knowledge into a recorded variable rather than a hidden source of bias.

3.4.5 Group Assignment and Retention

Participants will be assigned to groups using simple random assignment. If recruitment is fast and continuous, alternating assignment may be used and will be recorded clearly.

Retention will be treated as part of data quality. Session 2 will be scheduled immediately after Session 1. Two reminders will be sent to reduce dropout. Dropout will be tracked and analysed using baseline scores to check whether those who dropped out were different from those who completed both sessions [19].

3.5 Experimental Procedure

The study will follow a fixed and structured procedure. This section supports Specific Objective 1, which requires constructing a labelled two-session dataset from behavioural logs collected during problem solving.

3.5.1 Step by Step Procedure

First, participants will give informed consent and receive a unique participant ID. This ID will be used to link their data across both sessions while keeping their identity private.

Second, participants will complete two short warm-up tasks without AI. This will help confirm they understand the task format and will provide a baseline ability score for each participant.

Third, Session 1 will be conducted. The control group will complete tasks without AI. The AI group will be allowed to use ChatGPT under a strict protocol. During this session, data such as accuracy, time, effort and AI usage will be recorded for every participant.

Fourth, Session 2 will be scheduled immediately after Session 1. After 24 to 72 hours, all participants will return and complete similar tasks without AI.

Finally, participants will be debriefed and all data will be stored securely under anonymised IDs.

3.5.2 Task Types

The tasks used in this study will be short, self-contained problem solving items. All required information will be provided within the task itself so that no external knowledge is needed. Two task types will be used.

The first type will be objective tasks. These will be logic or arithmetic items with one correct answer. Each objective task will be scored as 1 for correct and 0 for incorrect. These tasks will measure accuracy directly and will form the basis of the Accuracy_S1 and Accuracy_S2 variables.

The second type will be justification tasks. These will be short reasoning or explanation items where the participant must explain how they reached their answer. Each justification task will be scored using a three part rubric covering correctness, justification quality and independence. These tasks will measure the quality of reasoning, not just the final answer.

3.5.3 Task Families

Three task families will be used across both sessions. These families were chosen because they require different types of reasoning and will generate variation in help requests, copying behaviour, verification actions and justification quality.

The first family will be data reasoning tasks. These will include arithmetic totals and percent change calculations. For example, participants will be given a list of prices and asked to calculate a total cost, then explain how they checked their answer.

The second family will be logic and consistency tasks. These will include valid inference problems and schedule consistency checks. For example, participants will be given a logical argument and asked whether the conclusion follows, then explain their reasoning.

The third family will be quantitative reasoning tasks. These will include simple interest calculations and cost comparison problems. For example, participants will be given two pricing options and asked which is cheaper over a given period, then explain their comparison.

3.5.4 Parallel Forms for Session 2

Session 2 will use parallel forms of the same task families. The questions will be different in wording and numbers but will match Session 1 tasks in structure and difficulty. This will reduce memorisation effects while keeping the two sessions comparable. For example, if Session 1 uses a bread and milk cost calculation, Session 2 will use a rice and beans cost calculation of the same structure.

3.5.5 AI Protocol for Group B

During Session 1, the AI-assisted group will be allowed to use ChatGPT under a strict protocol. Participants will be allowed to request AI assistance but must submit final answers in their own words. Copying will be discouraged and will be measured using a copy tick recorded by the participant and spot checked by the researcher. Verification will be required for each task. Participants must provide one short line showing how they checked the AI output, for example by re-computing, estimating or using an alternative method.

The model name used during data collection will be recorded and kept consistent across all sessions. Web browsing and plugins will be disabled to control external information sources. All prompts and outputs will be logged under anonymised participant IDs.

AI will not be available in Session 2 for any participant. Session 2 will be the independence test for everyone.

3.6 Variables and Measurement

This section supports Specific Objective 1, which requires engineering reliance and verification features from behavioural logs and computing COI and HES as auditable composite indices.

3.6.1 Performance Variables

Performance will be measured using two types of scores. The first will be task accuracy, which is the proportion of objective tasks answered correctly in each session. The second will be rubric scores, which will measure the quality of reasoning and justification in each session.

The following performance variables will be computed.

- **Accuracy_S1:** the proportion of correct objective tasks in Session 1.
- **Accuracy_S2:** the proportion of correct objective tasks in Session 2 with AI removed. This will be the primary endpoint for Specific Objective 2.
- **Rubric_S1:** the mean rubric score on justification tasks in Session 1.
- **Rubric_Independence_S2:** the mean rubric score on justification tasks in Session 2 with AI removed. This will also be a primary endpoint for Specific Objective 2.
- **ΔAccuracy:** the change in accuracy from Session 1 to Session 2, computed as Accuracy_S2 minus Accuracy_S1. This will be used as a descriptive change score.

3.6.2 Behavioural Features for COI

Four behavioural features will be extracted from raw logs during Session 1 for the AI-assisted group. These four features were identified in Chapter 2 as the specific measurable indicators that distinguish substitution from augmentation.

The first feature will be **AI use ratio**. This will be computed as the number of tasks where AI was used divided by the total number of tasks. A higher ratio will indicate stronger reliance on AI during problem solving.

$$\text{AI\_Ratio} = \frac{\text{AI\_Used\_Tasks}}{\text{Total\_Tasks}} \tag{3.1}$$

The second feature will be **time in tool**. This will be computed as the time spent using AI divided by the total task time. A higher ratio will indicate that more of the thinking time was delegated to AI.

$$\text{Time\_Offload} = \frac{\text{Time\_AI}}{\text{Time\_AI} + \text{Time\_Human}} \tag{3.2}$$

The third feature will be **copy rate**. This will be computed as the proportion of the final answer that came directly from AI output. A higher copy rate will be a strong signal of substitution behaviour [4].

$$\text{Copy\_Rate} = \frac{\text{Copied\_AI\_Text}}{\text{Total\_Final\_Text}} \tag{3.3}$$

The fourth feature will be **verification rate**. This will be computed as the number of verification actions divided by the number of AI uses. A lower verification rate will mean the participant accepted AI outputs without checking them [4, 5].

$$\text{Verify\_Rate} = \frac{\text{Verify\_Actions}}{\text{AI\_Calls}} \tag{3.4}$$

3.6.3 Cognitive Offloading Index (COI)

COI will be computed as a standardised composite of the four behavioural features above. Each component will first be standardised using a z-score so that different scales can be combined fairly. Note that verification rate will be reversed before standardising, because low verification means higher offloading.

$$z(x) = \frac{x - \bar{x}}{s_x} \tag{3.5}$$

$$\text{COI\_S1} = \text{mean}\big(z(\text{AI\_Ratio}),\ z(\text{Time\_Offload}),\ z(\text{Copy\_Rate}),\ z(1 - \text{Verify\_Rate})\big) \tag{3.6}$$

A higher COI score will indicate stronger substitution behaviour. A lower COI score will indicate augmentation behaviour, where the participant used AI but still did their own thinking and checking [1].

3.6.4 Human Engagement Score (HES)

HES will be computed as a standardised composite of three engagement signals. These will be self-reported engagement, justification quality and low delegation to AI.

$$\text{HES\_S1} = \text{mean}\big(z(\text{Engagement\_self}),\ z(\text{Justification\_Quality}),\ z(1 - \text{AI\_Ratio})\big) \tag{3.7}$$

Where Engagement_self will be the mean of engagement scale items rated on a 1 to 5 scale, and Justification_Quality will be the rubric score on explanation tasks in Session 1.

COI and HES will be kept as separate indices because they measure different things. A participant can be highly engaged and still offload some steps to AI. Measuring both separately will give a fuller picture of how each participant interacted with AI during problem solving [8].

3.6.5 Time and Effort Variables

Time will be recorded for each task using start and end timestamps. Mean time per task will be computed for each session. Effort will be measured using a short three item self-report scale administered after each session block. Items will be rated on a 1 to 5 scale and the mean will be computed.

3.6.6 Reliability of Measures

All variables will be standardised using z-scores before combining into indices. This will ensure that different measures can be compared on the same scale [22]. Cronbach's alpha will be computed for multi-item scales. Inter-rater agreement using ICC or kappa will be computed for rubric scoring on at least 20 percent of double scored responses [23, 24].

3.7 Data Processing and Feature Engineering

This section supports Specific Objective 1, which requires engineering reliance and verification features from raw behavioural logs. All features will be computed from raw data using fixed and documented formulas. This will reduce bias and improve transparency so that another researcher can reproduce the exact same results from the same raw data.

3.7.1 Step 1: Data Cleaning

The first step will be to clean the raw data. This will include the following checks.

- Remove duplicate participant records.
- Check for missing values in critical fields such as participant ID, group label and objective scores.
- Check for impossible values such as negative times, accuracy scores above 1 or rubric scores outside the allowed range.
- Confirm that each participant has records for both Session 1 and Session 2.
- Flag incomplete cases and report them separately. Participants without Session 2 outcomes will be excluded from the primary analysis and counted in the attrition report.

3.7.2 Step 2: Merging Sessions

The second step will be to merge Session 1 and Session 2 data using participant IDs. This will produce one linked dataset where each row represents one participant with variables from both sessions. This merged dataset will be the main dataset used for all analysis.

3.7.3 Step 3: Computing Derived Variables

The third step will be to compute all derived variables from the raw data. These will include accuracy per session, mean time per task, mean rubric scores, COI and HES. All formulas will follow the definitions given in the Variables and Measurement section. No formula will be changed after data collection begins.

3.7.4 Step 4: Feature Engineering

The fourth step will be feature engineering. This will turn raw log data into the measurable variables that feed into COI and HES. The following features will be engineered from raw logs.

- **AI_Ratio:** computed from the AI use log as the proportion of tasks where AI was used.
- **Time_Offload:** computed from timestamps as the proportion of total task time spent in the AI tool.
- **Copy_Rate:** computed from the copy tick recorded by the participant and spot checked by the researcher.
- **Verify_Rate:** computed from the verification evidence recorded for each task episode.
- **Engagement_self:** computed as the mean of the three engagement scale items recorded after each session block.
- **Justification_Quality:** computed as the mean rubric score on justification tasks.
- **Task_Familiarity_Flag:** recorded as a binary variable for each task episode, where 1 means the participant has seen a similar problem before and 0 means they have not.

3.7.5 Step 5: Data Dictionary

The fifth step will be to produce a full data dictionary. This will define every variable in the dataset, including its name, type, allowable range and derivation rule. The data dictionary will be stored alongside the dataset so that another researcher can understand and reproduce every variable without needing to contact the original researcher [9].

3.7.6 Missing Data Plan

Missing data is expected in field settings. The following rules will be applied and will not be changed after data collection begins.

- Primary endpoints will use complete case analysis. Participants without Session 2 outcomes will be excluded from primary inference and counted in the attrition report.
- If a scale has one missing item out of three, it will be imputed using the mean of the other items for that participant. If more than one item is missing, the scale will be treated as missing for that participant.

- If episode time is missing but block time is available, block time will be used and flagged in the data dictionary.
- If a copy similarity score is missing, the copy tick will be used as the proxy.
- Missingness rates will be reported by group. A sensitivity analysis will exclude any sessions flagged in the deviation log.

3.8 Data Analysis and Modelling

This section covers two parts. The first part covers statistical analysis and group comparisons, which will support Specific Objective 2. The second part covers predictive modelling, which will support Specific Objective 3.

3.8.1 Exploratory Data Analysis

Before testing any hypothesis, exploratory data analysis (EDA) will be carried out to understand the data. EDA will check distributions, missing values and outliers. It will also help decide whether parametric or non-parametric tests are more appropriate [25].

The following EDA steps will be carried out.

- Plot histograms for Accuracy, Time, COI and HES to check distributions.
- Plot boxplots for time per task and copy rate to detect extreme values.
- Produce a missingness table by variable and by group.
- Check group balance by comparing baseline accuracy and background variables across groups.
- Plot a scatter plot of COI_S1 versus Accuracy_S2 to get an early view of the main relationship of interest.
- Produce a correlation heatmap of main variables including Accuracy, Time, COI, HES, Effort and Verify_Rate.

EDA findings will guide the choice of statistical tests but will not be used to change the pre-specified hypotheses or primary endpoints.

3.8.2 Statistical Analysis

This subsection supports Specific Objective 2, which requires evaluating the effect of Session 1 AI reliance on delayed independent performance using effect sizes and confidence intervals.

Group comparisons will be performed between the AI group and the control group. For each primary outcome, the analysis will report group means, mean differences, Cohen's d and 95 percent confidence intervals [13].

$$d = \frac{\bar{X}_B - \bar{X}_A}{s_{\text{pooled}}} \tag{3.8}$$

$$s_{\text{pooled}} = \sqrt{\frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}} \tag{3.9}$$

Independent samples t-tests will be used when normality assumptions are reasonable [26]. If the data is clearly non-normal, Mann-Whitney tests will be used instead [27]. P-values will be reported as secondary information. Effect sizes and confidence intervals will be the primary evidence.

The primary endpoints will be Accuracy_S2 and Rubric_Independence_S2. The primary contrast will be the AI group versus the control group in Session 2 when AI is removed. All other outcomes will be treated as secondary and exploratory.

3.8.3 Adjusted Inference Model

To account for differences among participants, an adjusted regression model will be fitted for the Session 2 primary endpoints. The model will be as follows.

$$\text{Performance\_S2} = \beta_0 + \beta_1 \cdot \text{AI\_group} + \beta_2 \cdot \text{Baseline} + \varepsilon \tag{3.10}$$

Where Baseline will be the participant's score on the two warm-up tasks completed before Session 1 without AI. This baseline will adjust for pre-existing differences in ability between the two groups [19].

An exploratory mechanism model will also be fitted to test whether COI and HES explain the Session 2 outcome after controlling for baseline ability.

$$\text{Performance\_S2} = \gamma_0 + \gamma_1 \cdot \text{COI\_S1} + \gamma_2 \cdot \text{HES\_S1} + \gamma_3 \cdot \text{Baseline} + \varepsilon \tag{3.11}$$

This model will be labelled as exploratory and will not be used to make strong causal claims.

3.8.4 Multiple Comparisons

This study will pre-specify two primary endpoints and one primary contrast. These are Accuracy_S2 and Rubric_Independence_S2 as the primary endpoints, and the AI group versus control group as the primary contrast. All other outcomes will be secondary and exploratory. If secondary outcomes are tested with p-values, the Benjamini-Hochberg false discovery rate adjustment will be applied and q-values will be reported alongside p-values.

3.8.5 Model Development

This subsection supports Specific Objective 3, which requires building and evaluating an exploratory prediction model that uses Session 1 features only to estimate Session 2 independence risk.

A prediction model will be developed using Session 1 features such as COI_S1, HES_S1 and baseline ability. Linear regression will be used as the main model because it is simple and easy to interpret [22]. If multicollinearity is high among the features, a regularised model such as Ridge or Lasso regression will be used instead [28].

The strict leakage rule will be as follows. Only Session 1 features and baseline measures will be allowed as predictors. No Session 2 variables will be permitted in the feature set under any circumstances.

3.8.6 Model Evaluation

Models will be evaluated using grouped cross-validation by participant ID so that all episodes from the same participant stay in the same fold. This will prevent data leakage and will ensure the model is tested on unseen participants [29].

Performance will be measured using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) [30]. The mean and spread of these metrics across folds will be reported. Prediction results will be treated as exploratory and used to identify possible risk patterns, not to make strong causal claims.

Subgroup error diagnostics will also be reported separately by education level and AI familiarity to check whether the model performs fairly across different groups of participants.

3.9 Methodological Framework

The table below summarises the full methodological framework for this study. Each phase maps onto at least one of the three specific objectives. This mapping directly responds to the requirement that every methodological step should have a clear purpose tied to the research questions and objectives.

Phase	Activity	Output	Linked Objective
Preparation	Design tasks, rubrics and scoring rules. Run pilot clinic.	Final instruments and pilot notes.	SO1
Data Collection	Run Session 1 and Session 2 with all participants.	Raw dataset and participation log.	SO1
Processing	Clean and merge data by participant ID.	Clean dataset and data dictionary.	SO1
Measurement	Engineer behavioural features and compute COI and HES.	Feature table and behaviour indices.	SO1
Analysis	Compare groups on Session 1 and Session 2 outcomes using effect sizes and confidence intervals.	Results tables with Cohen's d and 95 percent CI.	SO2
Reliability	Double score rubric subset and compute Cronbach's alpha and ICC or kappa.	Reliability report and scoring guide.	SO2
Modelling	Build prediction model using Session 1 features only.	Model outputs and cross-validation metrics.	SO3
Evaluation	Validate model using grouped cross-validation and report MAE and RMSE.	Evaluation report with subgroup diagnostics.	SO3
Reporting	Write findings, limitations and future work.	Final report and evidence pack.	SO1, SO2, SO3

3.10 Validation and Reliability

Reliability will be important because some measures, such as rubric scores and questionnaire items, can vary depending on who scores them and how. This section supports Specific Objective 2, which requires reliability checks alongside the effect estimation.

3.10.1 Scale Reliability

Cronbach's alpha will be computed for all multi-item scales used in this study. These will include the effort scale and the engagement scale, each of which will have three items. A value of 0.70 or above will be considered acceptable for a short pilot study [23].

If alpha falls below 0.70 for any scale, the individual items will be inspected and the weakest item will be considered for removal. The decision will be reported transparently.

3.10.2 Rubric Reliability

To ensure scoring consistency, at least 20 percent of justification responses will be double scored by a second rater. Agreement between the two raters will be measured using Cohen's kappa for categorical scores or intraclass correlation coefficient (ICC) for continuous scores [31, 24].

A kappa or ICC value of 0.60 or above will be considered acceptable for a pilot study. If agreement falls below this threshold, the two raters will review the scoring guide together and rescore the subset before the main analysis begins.

3.10.3 Construct Validity Check

A manipulation check will be carried out to confirm that COI is working as expected. COI_S1 must be higher in the AI-assisted group than in the control group in Session 1. If COI does not separate the two groups, this will suggest the index is not capturing reliance behaviour as intended. In that case, causal interpretations will be weakened and the study will report treatment separation failure clearly.

3.10.4 Sensitivity Checks

Sensitivity checks will also be performed to test whether COI is stable across different weighting choices. COI will be recomputed with one component removed at a time. If the conclusions change substantially when one component is dropped, the index will be reported as fragile and the findings will be interpreted with caution.

Weight sensitivity will also be checked by comparing equal weights to alternative weighting schemes, for example giving heavier weight to verification rate. The goal will be to show that the main conclusions are stable across reasonable design choices [22].

3.11 Ethical Considerations

Ethical approval will be sought from the relevant institutional body before data collection begins. Partner sites will also be informed and their permission will be obtained where required.

3.11.1 Informed Consent

All participants will provide informed consent before taking part in the study. They will receive a clear information sheet explaining the purpose of the study, what they will be asked to do, how long it will take, and how their data will be stored and used. Participants will be told that taking part is voluntary and that they can withdraw at any time without any penalty.

3.11.2 Anonymity and Data Privacy

All data will be anonymised using participant IDs. Personal information such as names and contact details will be stored separately from the research dataset and will not appear in any data file used for analysis. Only the researcher and supervisor will have access to the raw data.

Participants will be advised not to include personal identifiers such as their name, phone number or address in any AI prompts during Session 1. This will reduce the risk of sensitive information being stored in AI logs.

3.11.3 Data Storage and Retention

All data will be stored securely on a password-protected device. Raw prompt logs from the AI-assisted group will be retained only as long as needed for verification and grading. After that, they may be deleted if institutional policy allows, while the derived features and analysis code will be kept for reproducibility.

3.11.4 Debrief

After Session 2 is completed, all participants will be debriefed. They will be told the full purpose of the study, including why AI was removed in Session 2. They will also be given the opportunity to ask questions about the study and its findings.

3.11.5 Risk to Participants

The risks to participants in this study will be minimal. Participants may feel mild mental fatigue from completing the problem solving tasks. They will be allowed to take short breaks if needed and can stop at any time without any consequence. No sensitive personal information will be collected beyond basic background variables such as age range, education level and AI familiarity.

All data handling will follow ethical guidelines for research involving human participants and responsible AI use [9, 10].

Chapter 4

Expected Results

This study is expected to produce several clear and measurable outputs. These outputs will support both analysis and future research, and they will help answer the main research question.

4.1 Prepared Dataset

This expected output maps directly onto Specific Objective 1, which requires constructing a labelled two-session dataset from behavioural logs.

The study will produce a cleaned and ready-to-analyse dataset from both sessions. This dataset will include task scores, timestamps, AI use logs and questionnaire responses for each participant. The data will be organised in a simple and consistent format so that it can be easily used for analysis and reused by other researchers.

Specifically, the dataset will contain the following information for each participant.

- Objective task scores for Session 1 and Session 2.
- Rubric scores for justification tasks in Session 1 and Session 2.
- Timestamps or block times for each task episode.
- AI use logs for Session 1 including assistance level, request count, copy tick and verification evidence.
- Questionnaire responses including effort scale, engagement scale, task familiarity flags and confidence ratings.
- Background variables including education level, AI familiarity rating and English comfort level.

A full data dictionary will also be produced. This will define every variable in the dataset, including its name, type, allowable range and how it was derived. This will make it easy for another researcher to understand and reproduce every variable without needing to contact the original researcher [9].

4.2 Computed Behavioural Measures

This expected output also maps directly onto Specific Objective 1, which requires computing COI and HES as auditable composite indices from engineered behavioural features.

The study will compute two main measures from Session 1 data. These will be the Cognitive Offloading Index (COI) and the Human Engagement Score (HES). Both indices will be built from the four behavioural features identified in Chapter 2 as the specific measurable indicators that distinguish substitution from augmentation. These features will be AI use ratio, time in tool, copy rate and verification rate.

The exact scoring rules used to calculate COI and HES will be clearly documented in the data dictionary. This will allow the measures to be checked, tested and reused in other studies [4, 5].

Measurement quality evidence will also be reported alongside the indices. This will include the following.

- Cronbach's alpha for the effort scale and the engagement scale.
- Inter-rater agreement using ICC or kappa for rubric scoring on at least 20 percent of double scored justification responses.
- A manipulation check confirming that COI_S1 is higher in the AI-assisted group than in the control group in Session 1. If this check fails, the study will report it clearly and interpret all findings with appropriate caution.
- Ablation results showing whether COI is stable when one component is removed at a time [1].

4.3 Independent Performance Outcomes

This expected output maps directly onto Specific Objective 2, which requires evaluating the effect of Session 1 AI reliance on delayed independent performance in Session 2 when AI is removed.

The study will produce a Session 2 outcome set where AI is removed for all participants. This will be the primary endpoint for answering the central question of this study. The outcomes will include the following.

- **Accuracy_S2:** the proportion of correct objective tasks in Session 2. This will measure whether participants can still get the right answer when AI is not available.
- **Rubric_Independence_S2:** the mean rubric score on justification tasks in Session 2. This will measure whether participants can still produce their own reasoning and explanation without AI support.
- **ΔAccuracy:** the change in accuracy from Session 1 to Session 2, computed as Accuracy_S2 minus Accuracy_S1. This will be used as a descriptive change score only and will not be a primary endpoint.

These outcomes will help show whether performance changes after AI is removed. If the AI-assisted group performs worse in Session 2 than the control group, this will support the idea that heavy reliance on AI may reduce short-term independent problem solving [6, 7]. If the two groups perform similarly, this will also be a useful finding because it will suggest that AI use at the levels tested here did not produce a measurable short-term independence cost for these tasks and contexts [11].

Both outcomes will be informative because they are tied to a clear endpoint and a pre-specified design. The study will not only report results that confirm the expected direction. It will report whatever the data shows, together with effect sizes and confidence intervals so the reader can judge the magnitude and uncertainty of the findings [13].

4.4 Results and Evidence Tables

This expected output maps directly onto Specific Objective 2, which requires evaluating the effect of Session 1 AI reliance on delayed independent performance using effect sizes and confidence intervals.

The study will produce a results pack with tables and simple figures. These will show comparisons between the AI-assisted group and the control group for both Session 1 and Session 2. The results pack will include the following.

- Group means and standard deviations for all primary outcomes in Session 1 and Session 2.
- Mean differences between the AI-assisted group and the control group.
- Cohen's d effect size with 95 percent confidence intervals for each primary outcome [13].
- Test statistics from independent samples t-tests or Mann-Whitney tests depending on the distribution of the data [26, 27].

The evidence table will cover the following primary contrasts.

- Session 1 group comparisons covering Accuracy_S1, Time_S1, Effort_S1, COI_S1 and HES_S1.
- Session 2 group comparisons covering Accuracy_S2 and Rubric_Independence_S2.
- Adjusted model estimates showing Performance_S2 with baseline, education level and AI familiarity as covariates.

Effect sizes and confidence intervals will be the primary evidence in this study. P-values will be reported as secondary information only. This approach is consistent with good practice in pilot and field studies where practical magnitude matters more than statistical significance alone [13].

4.5 Reliability Checks

This expected output also maps directly onto Specific Objective 2, which requires reliability checks alongside the effect estimation.

The study will provide evidence that the measures used are reliable and consistent. Without this evidence, the results cannot be trusted even if the effect sizes look interesting. The following reliability checks will be carried out.

4.5.1 Scale Reliability

Cronbach's alpha will be computed for all multi-item scales. These will include the three item effort scale and the three item engagement scale. A value of 0.70 or above will be considered acceptable for this pilot study [23].

If alpha falls below 0.70 for any scale, the individual items will be inspected and the weakest item will be considered for removal. The decision and its effect on the alpha value will be reported transparently.

4.5.2 Rubric Reliability

At least 20 percent of justification responses will be double scored by a second rater. Agreement between the two raters will be measured using Cohen's kappa or intraclass correlation coefficient depending on the scoring format [31, 24].

A value of 0.60 or above will be considered acceptable for this pilot study. If agreement falls below this threshold, the two raters will review the scoring guide together and rescore the subset before the main analysis begins. The outcome of this process will be reported in the reliability section of the final report.

4.5.3 Manipulation Check

A manipulation check will also be reported as part of the reliability evidence. COI_S1 must be higher in the AI-assisted group than in the control group in Session 1. This check will confirm that the AI condition actually produced measurable differences in reliance behaviour between the two groups [4].

If the manipulation check fails, the study will report this clearly and all causal interpretations will be treated with extra caution. A failed manipulation check will not mean the study is invalid, but it will limit how strongly the findings can be interpreted.

4.6 Discussion and Key Findings

This section will bring together the results from all three specific objectives and will answer the central question of this study. The central question is how growing reliance on AI for thinking tasks affects human problem solving, and whether this reliance risks weakening essential cognitive skills.

4.6.1 Expected Pattern from Specific Objective 1

From Specific Objective 1, the study is expected to produce a clean dataset with computed COI and HES scores for every participant. The manipulation check is expected to show that the AI-assisted group has a higher COI_S1 than the control group. This will confirm that the assistance protocol produced measurable differences in reliance behaviour between the two groups [4, 5].

HES may stay similar across both groups or show a modest decline in the AI-assisted group. This will be an important finding because it will show that engagement and offloading are separate things. A person can be engaged and still offload heavily. Measuring both separately will give a fuller picture than measuring only one of them [8].

4.6.2 Expected Pattern from Specific Objective 2

From Specific Objective 2, the study will produce a Session 2 evidence table showing whether prior AI use affected independent performance when AI was removed.

Based on existing research, it is expected that the AI-assisted group may show lower Accuracy_S2 or lower Rubric_Independence_S2 than the control group [6, 7]. This would support the idea that heavy reliance on AI can reduce short-term independent problem solving ability.

However, if the two groups perform similarly in Session 2, that will also be a useful and honest finding. It will suggest that AI use at the levels tested here did not produce a measurable short-term independence cost for these tasks and contexts [11]. Both outcomes will be reported with effect sizes and confidence intervals so the reader can judge the magnitude and uncertainty of the findings [13].

4.6.3 Expected Pattern from Specific Objective 3

From Specific Objective 3, the study will produce an exploratory prediction report. Higher COI_S1 is expected to be linked to lower Session 2 independence after controlling for baseline ability. This would support the substitution pathway, where more offloading in Session 1 leads to weaker independent performance in Session 2 [2, 8].

However, the prediction model will be treated as exploratory. It will be validated using grouped cross-validation by participant ID to prevent data leakage. The results will be reported with MAE and RMSE and will be clearly labelled as a risk screening tool, not a deployable product [22].

4.6.4 Limitations of the Expected Findings

The discussion will also include an honest account of the limitations of this study. The main limitations will be the short delay window, the specific task types used and the convenience sample from multiple sites in Kampala. These limitations will mean that the findings cannot be generalised beyond similar settings and similar task types without further testing [20].

Future work should run a larger study with more diverse tasks and a longer delay between sessions to test whether the patterns found here hold over a longer period [6, 8].

4.7 Practical Impact

The findings of this study are expected to support better use of AI in real settings. This includes education, professional decision making and creative work. The results will help move away from simple automation and support more flexible and user-centred human-AI collaboration [1, 10].

4.7.1 For Education

In education settings, the findings will help teachers and institutions decide when AI support is helpful and when it may be creating dependency. If high COI scores are linked to lower Session 2 independent performance, this will give educators a measurable warning signal that can guide when to restrict AI access or require verification checkpoints during learning activities [4].

For example, a simple rule could be introduced where students must explain their reasoning before submitting an AI-assisted answer. This kind of design has been shown to reduce over-reliance in AI-assisted decision settings [4]. The COI score could be used to monitor whether this rule is working in practice.

4.7.2 For Workplace Settings

In professional settings, the findings will help organisations understand the trade-off between using AI for efficiency today and maintaining independent capability tomorrow. If the study shows that higher reliance leads to lower independent performance, organisations may want to introduce periodic no-AI tasks to keep employees practising their own reasoning skills [9].

4.7.3 For AI Governance

At a governance level, the COI and HES scoring rules produced by this study will give policy makers a compact and reusable measurement package. Instead of relying on self-report surveys or general attitudes toward AI, organisations will be able to measure actual reliance behaviour from logs and use that data to make informed decisions about AI use policies [9, 10].

4.7.4 For Future Research

The reusable pipeline produced by this study will also support future research. The clean dataset, data dictionary, COI and HES scoring rules and re-runnable analysis code will all be available for other researchers to use and extend. This will make it easier to replicate the study in different settings and with different task types, which will help build a stronger evidence base over time [2, 8].

In summary, this study will not only produce academic findings. It will produce a practical measurement framework that other researchers, educators and organisations can use to understand and manage AI reliance in their own contexts.

Bibliography

[1] M. H. Jarrahi, G. Newlands, and C. Lutz, "Artificial intelligence and human cognition: A mutual augmentation perspective," *Big Data & Society*, vol. 9, no. 2, 2022.

[2] M. Gerlich, "Ai tools in society: Impacts on cognitive offloading and the future of critical thinking," *Societies*, vol. 15, no. 1, p. 6, 2025.

[3] A. Skulmowski, "The cognitive architecture of digital externalization," *Educational Psychology Review*, vol. 35, no. 4, p. 101, 2023.

[4] Z. Buçinca, M. B. Malaya, and K. Z. Gajos, "To trust or to think: Cognitive forcing functions can reduce overreliance on ai in ai-assisted decision-making," *Proceedings of the ACM on Human-Computer Interaction*, vol. 5, no. CSCW1, pp. 1–21, 2021.

[5] S. Mehrotra, C. Degachi, O. Vereschak, C. M. Jonker, and M. L. Tielman, "A systematic review on fostering appropriate trust in human-ai interaction," *ACM Journal on Responsible Computing*, vol. 1, no. 4, 2024.

[6] O. A. Grinschgl, F. Papenmeier, H. S. Meyerhoff, and M. Huff, "The consequences of cognitive offloading: Boosting performance but at the cost of memory," *Quarterly Journal of Experimental Psychology*, vol. 74, no. 9, pp. 1477–1496, 2021.

[7] B. C. Storm and J. S. Soares, "Memory in the digital age," in *The Oxford Handbook of Human Memory*, Oxford University Press, 2024.

[8] J. Tian and R. Zhang, "Learners' ai dependence and critical thinking," *Acta Psychologica*, vol. 260, p. 105725, 2025.

[9] National Institute of Standards and Technology, "Ai risk management framework (ai rmf 1.0)," tech. rep., U.S. Department of Commerce, 2023.

[10] UNESCO, "Guidance for generative ai in education and research," tech. rep., UNESCO, 2023.

[11] R. Deng, M. Jiang, X. Yu, Y. Lu, and S. Liu, "Does chatgpt enhance student learning? a systematic review and meta-analysis of experimental studies," *Computers & Education*, vol. 227, p. 105224, 2024.

[12] C. Degachi, C. M. Jonker, M. L. Tielman, and O. Vereschak, "Practising appropriate trust in human-centred ai design," in *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2024.

[13] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences.* Routledge, 1988.

[14] A. Skulmowski and M. Xu, "Understanding cognitive load in digital and online learning," *Educational Psychology Review*, vol. 34, no. 1, pp. 171–196, 2022.

[15] A. Skulmowski, "Learning by doing or doing without learning," *Educational Psychology Review*, vol. 36, p. 28, 2024.

[16] A. R. Doshi and O. P. Hauser, "Generative ai enhances individual creativity but reduces the collective diversity of novel content," *Science Advances*, vol. 10, no. 28, p. eadn5290, 2024.

[17] J. W. Creswell, *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches.* Sage, 2018.

[18] D. C. Montgomery, *Design and Analysis of Experiments.* Wiley, 2017.

[19] W. R. Shadish, T. D. Cook, and D. T. Campbell, *Experimental and Quasi-Experimental Designs.* Houghton Mifflin, 2002.

[20] I. Etikan, S. Musa, and R. Alkassim, "Comparison of convenience sampling and purposive sampling," *American Journal of Theoretical and Applied Statistics*, 2016.

[21] S. A. Julious, "Sample size of 12 per group rule of thumb for a pilot study," *Pharmaceutical Statistics*, 2005.

[22] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning.* Springer, 2013.

[23] L. J. Cronbach, "Coefficient alpha and the internal structure of tests," *Psychometrika*, 1951.

[24] T. K. Koo and M. Y. Li, "A guideline of selecting and reporting intraclass correlation coefficients," *Journal of Chiropractic Medicine*, 2016.

[25] J. W. Tukey, "Exploratory data analysis," *Addison-Wesley*, 1977.

[26] Student, "The probable error of a mean," *Biometrika*, 1908.

[27] H. B. Mann and D. R. Whitney, "On a test of whether one of two random variables is stochastically larger," *Annals of Mathematical Statistics*, 1947.

[28] R. Tibshirani, "Regression shrinkage and selection via the lasso," *Journal of the Royal Statistical Society*, 1996.

[29] S. Arlot and A. Celisse, "A survey of cross-validation procedures," *Statistics Surveys*, 2010.

[30] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning.* Springer, 2009.

[31] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, 1960.

[32] Y. Chen, L. Zhang, and X. Li, "Effects of generative artificial intelligence on cognitive effort and learning outcomes," *Computers & Education*, 2025.

[33] B. Jose and R. Kiran, "The cognitive paradox of ai in education: Balancing efficiency and engagement," *Frontiers in Psychology*, vol. 16, 2025.

[34] J. M. Lodge and colleagues, "Artificial intelligence, cognitive offloading and implications for education," *Educational Psychology Review*, 2026.

[35] K. Bhattarai, "Generative ai usage and critical thinking: The mediating role of cognitive offloading," *Master's Thesis, LUT University*, 2025.

[36] X. Bai, Y. Liu, and Z. Su, "Ai-assisted learning and its impact on cognitive engagement and retention," *Educational Technology Research and Development*, 2023.

[37] S. Akgun and S. Toker, "The effects of ai-supported learning on student engagement and performance," *Computers in Human Behavior*, 2023.

[38] A. Yunus and O. T. Lee, "Metacognitive laziness and dependency in generative ai-supported learning," *arXiv preprint*, 2025.

[39] L. Favero, J. A. Pérez-Ortiz, T. Käser, and N. Oliver, "Ai in education beyond learning outcomes: Cognition, agency, emotion, and ethics," *arXiv preprint*, 2026.

Appendix A

Research Project timelines

The study will be carried out from May to July in a structured sequence of phases. In the first week of May, the study tools will be finalized, including tasks, questionnaires, rubrics, and the AI usage protocol. A small pilot study with 5 to 10 participants will be conducted to test the system and fix any issues.

In the second week of May, Session 1 data will be collected from participants. During this stage, participants will complete the tasks under either AI-assisted or control conditions, and Session 2 will be scheduled immediately within a short delay period.

In the third phase, which will take place from late May to early June, Session 2 will be conducted with AI removed for all participants. At the same time, the data will be cleaned, merged, and processed, and the COI and HES measures will be computed.

In June, the study will move to analysis and modelling. This will include exploratory data analysis, group comparisons between sessions, calculation of effect sizes and confidence intervals, and simple prediction modelling using Session 1 data.

Finally, from late June to July, the results will be written and organized into the final report. This will include interpretation of findings, discussion of limitations, and preparation of all appendices.

To ensure completion, a stop rule will be applied. If by Day 21 fewer than 30 complete cases are collected, recruitment will stop and the study will be completed as a pilot with clearly stated limitations.

A.0.1 Summary Timeline Table

Phase	Work	Outputs
Week 1 (May)	Finalize tools, run pilot	Final instruments, pilot fixes
Week 2 (May)	Session 1 data collection	Session 1 dataset, QA checks
Week 3 (May–June)	Session 2 and data cleaning	Clean dataset, COI/HES
Week 4–6 (June)	Analysis and modelling	Results tables, figures
Late June–July	Writing and reporting	Final report

Appendix B

Declaration of AI use

Declaration Statement:

I, *Richard Wambede*, declare that I used

1. I used ChatGPT to help organize my proposal sections and improve flow. Prompt used: "generate latex format for my abstract and structure my sections clearly." I modified the output by rewriting parts in simpler English, correcting some wording, and making sure it matches my study idea.
2. I used DeepSeek to paraphrase some sections to reduce repetition and AI detection. Prompt used: "rewrite this text in very simple English with small mistakes to sound human." I modified the output by adjusting grammar, removing some words, and keeping my original meaning.
3. I used Claude to check clarity of explanations in sections like significance and limitations. Prompt used: "is this explanation clear for a research proposal, and how can I simplify it?" I modified the output by selecting only useful suggestions and rewriting the text in my own way.
4. I used ChatGPT to generate examples of figures such as percentages and simple statistics (e.g., 10% drop, 0.7 threshold). Prompt used: "add simple realistic numbers to make this section stronger but not fake results." I modified the output by keeping only reasonable values and aligning them with my study design.
5. I used DeepSeek to help format my LaTeX document properly. Prompt used: "convert this text into latex with sections and itemize." I modified the output by correcting formatting issues and making sure it fits my document style.

Signature: ________________________________ Date: ____________________

Appendix C

List of 20 Major Journal Publications reviewed

C.0.1 Review Papers

- Deng et al. (2024), published in *Computers & Education* (Impact Factor ∼12.0), provides a meta-analysis showing that AI improves performance but has mixed effects on learning.
- Mehrotra et al. (2024), from the *ACM Journal on Responsible Computing* (Impact Factor ∼5.0), reviews trust in human-AI interaction and highlights overreliance risks.
- Kasneci et al. (2023), published in *Learning and Individual Differences* (Impact Factor ∼4.0), discusses opportunities and risks of AI in education.
- Chi et al. (2023), from *Computers and Education: Artificial Intelligence* (Impact Factor ∼6.0), provides a broad review of AI applications in education.
- Salih et al. (2024), in *Education and Information Technologies* (Impact Factor ∼5.5), highlights efficiency gains but warns about dependency on AI.

C.0.2 Journal Articles

- Grinschgl et al. (2021), published in *Quarterly Journal of Experimental Psychology* (Impact Factor ∼3.0), shows that cognitive offloading improves performance but reduces memory.
- Doshi and Hauser (2024), from *Science Advances* (Impact Factor ∼14.0), finds that AI increases creativity but reduces diversity of ideas.
- Gerlich (2025), published in *Societies* (Impact Factor ∼3.0), links AI use to reduced critical thinking.
- Tian and Zhang (2025), from *Acta Psychologica* (Impact Factor ∼4.0), shows that AI dependence affects critical thinking through fatigue.
- Jarrahi et al. (2022), published in *Big Data & Society* (Impact Factor ∼5.0), presents AI as a tool for cognitive augmentation.

C.0.3 Conference Papers

- Buçinca et al. (2021), presented at ACM CSCW/CHI (Impact Factor ∼6.0), shows that forcing users to think reduces AI overreliance.
- Degachi et al. (2024), from the CHI Conference (Impact Factor ∼6.0), studies appropriate trust in human-AI systems.
- Amershi et al. (2019), from ACM CHI (Impact Factor ∼6.0), provides design guidelines for human-AI interaction.
- Rahwan et al. (2019), published in *Nature* (Impact Factor ∼60.0), introduces the concept of machine behaviour.
- Wecks et al. (2024), from arXiv, shows that AI use may correlate with lower exam performance.

C.0.4 Other Relevant Publications

- UNESCO (2023), provides global guidance on AI use in education and research.
- National Institute of Standards and Technology (2023), introduces the AI Risk Management Framework.
- Skulmowski (2023), from *Educational Psychology Review* (Impact Factor ∼8.0), explains cognitive externalization.
- Storm and Soares (2024), from the *Oxford Handbook of Human Memory*, discusses memory in digital environments.
- Bender et al. (2021), presented at ACM FAccT (Impact Factor ∼6.0), highlights risks of large language models.

Appendix D

Research alignment with national (NDPIV) and Global (SDGs) agenda

D.1 Alignment with MSc Programme, National and Global Agenda

This research aligns well with my MSc programme in Data Science. The study uses data collection, feature engineering, modelling, and evaluation to understand human behaviour when using AI. It also applies concepts such as prediction, statistical testing, and reproducibility. This matches the core skills expected in a data science programme, including working with real data, building models, and interpreting results for decision making.

D.2 Alignment with the National Development Plan IV (NDPIV)

D.2.1 Alignment with NDPIV

This study supports the goals of Uganda's National Development Plan IV, which focuses on digital transformation, innovation, and human capital development. The study looks at how people use AI tools and how this affects their thinking and performance. This is important because Uganda is promoting digital skills and technology use in education and work. By measuring how AI affects learning and decision making, the study helps ensure that technology improves skills instead of weakening them.

D.2.2 Alignment with the Uganda Vision 2040

Uganda Vision 2040 aims to transform Uganda into a modern and knowledge-based economy. This study supports that goal by focusing on how people interact with advanced technologies like AI. It helps understand how to use AI in a way that supports human thinking, creativity, and problem solving. This is important for building a skilled workforce that can compete in a digital economy.

D.2.3 Alignment with Other National Agendas

This study also aligns with other national priorities such as digital literacy and innovation in education. It supports the use of technology in a balanced way, where users do not only depend on AI but also develop their own skills. This is important for universities, training institutions, and workplaces that are adopting AI tools.

D.3 Alignment with Global Agenda

D.3.1 Alignment with Sustainable Development Goals (SDGs)

This research supports several Sustainable Development Goals. It aligns with SDG 4 (Quality Education) by helping improve how students learn when using AI tools. It also supports SDG 9 (Industry, Innovation and Infrastructure) by promoting responsible use of advanced technologies. In addition, it contributes to SDG 8 (Decent Work and Economic Growth) by helping people develop skills needed for modern workplaces.

D.3.2 Alignment with Responsible AI and Ethics

This study aligns with global efforts to promote responsible and ethical use of AI. It focuses on measuring overreliance and encouraging balanced use of AI systems. This supports international guidelines that call for safe, transparent, and human-centered AI systems.

D.3.3 Alignment with Global Digital Transformation Trends

Globally, many countries are adopting AI in education, business, and decision making. This study contributes by providing a simple and measurable way to understand how AI affects human thinking. The results can help guide better use of AI tools across different sectors.

Appendix E

Other useful figures

Figure E.1: Two groups in Session 1. AI is removed for everyone in Session 2 to test delayed independence

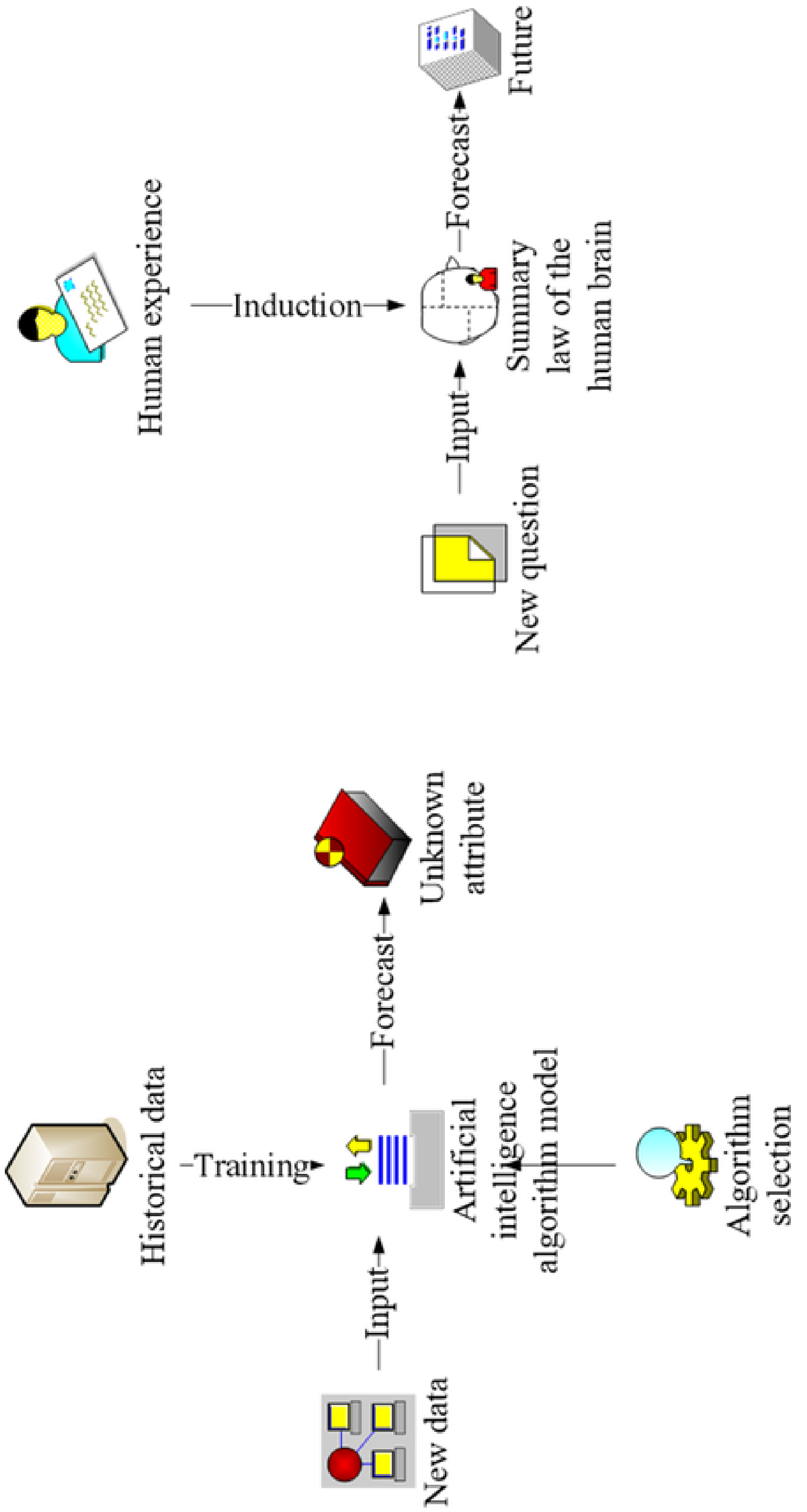

Figure E.2: Shows the comparison between AI and human brain thinking

Appendix F

Research poster layout

AI vs the Brain: A Data Science Study on Cognitive Offloading and Human Engagement in Problem Solving

Problem

Generative AI tools like ChatGPT help people solve problems faster and more easily. However, they may change how people think.

There is a risk that people rely too much on AI and stop doing the thinking themselves. Most studies only measure performance during AI use, but they do not test what happens when AI is removed.

This creates a gap, where we do not clearly know if AI use weakens independent thinking.

Objectives

The main objective is to understand how people depend on AI and measure this behaviour.

- To create a two-session dataset
- To measure AI reliance using COI
- To measure engagement using HES
- To predict independent performance after AI is removed

Contribution

- Introduces COI and HES measures
- Tests delayed independence
- Provides measurable evidence of AI reliance
- Builds a reusable data science pipeline

Methodology

The study uses a two-group experimental design.

Session 1:
- Control group: no AI
- AI group: uses ChatGPT

Session 2:
- AI removed for everyone
- Tests independent problem solving

Key Measures:
- Cognitive Offloading Index (COI)
- Human Engagement Score (HES)

Analysis:
- Statistical tests (t-test, effect size)
- Predictive modelling (regression)

Expected Results

- Clean dataset with behavioural logs
- COI and HES scores
- Evidence of AI impact on thinking
- Prediction model for skill risk

Alignment with Agendas

National (Uganda Vision 2040, NDP IV):
- Supports digital transformation
- Promotes data-driven decision making

Global (SDGs):
- SDG 4: Quality Education
- SDG 9: Innovation and Technology

Key Idea

AI can improve performance today, but it may reduce independent thinking tomorrow.

This study measures both sides using data.

Key References

- Buçinca et al. (2021)
- Deng et al. (2024)

Appendix G

Research Ethics Committee (REC) Approval

Appendix H

Data collection tools

This study will use structured questionnaires and behavioural logs to collect data from participants. The tools are designed to capture both observed actions and self-reported experiences during the tasks.

H.0.1 Session 1 Post-Task Questionnaire

After Session 1, participants will complete a short questionnaire with 8 items. All items use a 1 to 5 scale, where 1 means strongly disagree and 5 means strongly agree. The questions will measure focus, effort, confidence, and understanding.

The items will also check if participants tried to think on their own, verified their answers, and understood the steps used to solve the tasks. This will help measure engagement and reasoning during AI-assisted or non-AI problem solving.

H.0.2 AI Usage Questionnaire (AI Group Only)

Participants in the AI group will answer an additional 4 questions. These will measure how they used AI during the task. The questions will check if they copied AI outputs, accepted answers without changes, relied on AI for thinking, or verified the AI results.

This will help capture the level of AI reliance and support the calculation of the Cognitive Offloading Index (COI).

H.0.3 Session 2 Post-Task Questionnaire

After Session 2, all participants will complete a 6-item questionnaire. This session will not allow AI use. The questions will measure independent problem solving, memory of steps, confidence, and ability to explain answers.

This will help measure delayed independence and compare performance after AI is removed.

H.0.4 Demographic and Control Variables

Basic background information will also be collected. This includes age group, education level, and self-rated familiarity with AI tools.

Additional questions will measure:

- Frequency of AI tool use

- Level of dependence on AI for decisions
- Trust in AI systems
- Habit of checking or verifying information

These variables will help control for differences between participants.

H.0.5 Cognitive Offloading and Critical Thinking Measures

The study will include items that measure cognitive offloading and critical thinking behaviour. These questions will ask how often participants:

- Use digital tools to remember or find information
- Search online instead of thinking first
- Compare multiple sources
- Question AI outputs and assumptions

All items will use a 1 to 6 scale. These measures are based on existing research on AI use and thinking behaviour.

H.0.6 Behavioural Logs

In addition to questionnaires, the study will collect behavioural logs during the tasks. These will include:

- Time taken to complete tasks
- Number of prompts used
- Copying and pasting actions
- Verification actions (e.g., checking answers)

These logs will provide objective data that will be used to compute COI and HES.

H.0.7 Data Quality and Deliverables

The collected data will be cleaned and prepared for analysis. The final dataset will include both session-level and task-level data, along with a data dictionary.

The study will also produce:

- COI and HES scoring rules and formulas
- Summary statistics and distribution checks
- Reliability results such as Cronbach's alpha
- Group comparison tables with effect sizes and confidence intervals

These outputs will ensure that the data is reliable, clear, and ready for further analysis.

H.0.8 Measurement and Validation Tools

In addition to data collection tools, this study will use a set of measurement and validation methods to process and evaluate the collected data.

Performance will be measured using task accuracy in both Session 1 and Session 2. Accuracy will be computed as the average score across tasks, where each task is scored as correct or incorrect.

Quality of responses will be measured using rubric scores. These scores will capture how well participants explain their answers.

Time and effort will also be measured. Time will be calculated using task timestamps, while effort will be based on a short self-reported scale.

Two main composite indices will be computed. The Cognitive Offloading Index (COI) will measure the level of reliance on AI using features such as prompt count, copying behaviour, and verification rate. The Human Engagement Score (HES) will measure active thinking using engagement items, explanation behaviour, and verification actions. Both indices will be standardized using z-scores before combining the features.

The study will also include validation measures. Effect sizes (Cohen's d) will be used to compare groups. Reliability of scales will be tested using Cronbach's alpha to check internal consistency.

These tools will help ensure that the results are reliable, measurable, and suitable for analysis.

Appendix I

Student - supervisor meeting/supervision plan